STAT 709 Mathematical Statistics Lecture 21 - Fall 2026

Lecture 21: Hypothesis tests and p-values

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Report an issue

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Source and provenance. Most of the contents of the Part VII lectures are taken from Bickel and Doksum (BD), *Mathematical Statistics: Basic Ideas and Selected Topics*, Volume I, second edition [2], and adapted to the course notation. The Part VII lectures are newly prepared; they are **not the original lecture notes from previous years**. Additional explanations, examples, and figures are identified and credited below.

Lecture guide. This is the first of two lectures for Part VII: Testing and confidence sets. The proof of Theorem 1 is low priority, as assigned by the instructor. The other priority labels follow the course reading conventions.

Reading. The main reference is Bickel and Doksum (BD), Volume I, second edition (2001): Section 4.1, pp. 213–223; Section 4.2, pp. 223–227; and selected material from Section 4.3, pp. 227–230 [2]. Definitions and proofs are restated in the course notation; textbook numbers below refer to this edition. Additional examples and exercises are credited where used.

High priority *Overview*

An estimate answers "how large is the parameter?" A test asks whether the data give enough evidence against a specified claim. We control false rejection under the null, study power under the alternative, and use a p-value to locate the observed statistic in its null distribution. Sections 1–4 compare two fully specified distributions: simple versus simple hypotheses. They develop errors, a Normal example, p-values, and Neyman–Pearson. Section 5 then allows sets of distributions, explains uniform error control and UMP tests, and introduces generalized likelihood ratios.

Recap from earlier lectures

From Lectures 6, 8 and 10: earlier ideas used here

Reminder only: this block introduces no new exam requirements.

- **Tests as decisions**. A test maps data to reject or do not reject; risk averages a loss under a fixed model. Used in error control.
- **Likelihood**. With data fixed, the likelihood compares parameter values using their densities or masses. Used in the Neyman–Pearson lemma.
- **Normal approximation**. A centered, standardized sample mean is asymptotically Normal under the CLT assumptions. Used to distinguish exact from approximate calibration.

Learning goals

- Specify hypotheses, a test statistic, and a rejection region.
- Distinguish significance level, size, power, and the two error probabilities.
- Calculate and interpret one-sided and two-sided p-values.
- Use the Neyman–Pearson lemma and distinguish simple from generalized likelihood ratios; its proof is optional reading.

1 Hypotheses, decisions, and errors

High priority *From a scientific question to a statistical rule*

Start with two fully specified distributions for the data X:

$$H_0 : P = P_0 \qquad \text{against} \qquad H_1 : P = P_1.$$

The **null hypothesis** is the claim whose false rejection we will control. The **alternative hypothesis** describes the departures we want to detect. Choosing the model and the direction of the alternative is part of formulating the question, before examining the result.

Definition 1 (Simple and composite hypotheses; BD Section 4.1, p. 215)**.** *A hypothesis is **simple** if it specifies one distribution of the data, and **composite** if it allows more than one.*

For i.i.d. $N(\mu, \sigma^2)$ data with known $\sigma^2 > 0$, comparing $\mu = 0$ with $\mu = 1$ is a simple-versus-simple problem. Each hypothesis specifies one distribution of the whole sample. We first work in this setting; Section 5 develops composite hypotheses.

A nonrandomized test is a measurable function $\varphi : \mathcal{X} \to \{0, 1\}$. The decision $\varphi(X) = 1$ means reject H_0; $\varphi(X) = 0$ means do not reject. Its **rejection region** is $R = \{x : \varphi(x) = 1\}$. Often $R = \{x : T(x) \geq c\}$ for a statistic T and a critical value c. The statistic alone does not specify a test: its cutoff and rejection direction matter.

	H_0 true	H_1 true
Reject H_0	Type I error	Correct detection
Do not reject H_0	Correct nonrejection	Type II error

Definition 2 (Power, size, and level for simple hypotheses; BD Definition 4.1.1, specialized)**.** *For a test φ, its **size** and **power** are, respectively,*

$$P_0\{\varphi(X) = 1\}, \qquad P_1\{\varphi(X) = 1\}.$$

*The size is the Type I error probability. The Type II error probability is $\beta = P_1\{\varphi(X) = 0\} = 1 - \text{power}$. For a specified $0 < \alpha < 1$, the test has **level** α if its size is at most α.*

BD writes β for the power function; here π denotes power and β denotes Type II error. The size is a property of the rule; α is its chosen error budget. A size-0.03 test is a level-0.05 test. The two errors use different distributions: P_0 for false rejection and P_1 for missed detection. Their probabilities need not add to one.

Instructor's Note *A false-alarm budget*

Think of a smoke alarm. Lowering the trigger threshold helps detect small fires, but also produces more false alarms. The significance level sets a false-alarm budget. Among rules that respect that budget, we want an alarm that is sensitive to the fires we care about. An alarm that never rings has no false alarms and is still useless.

Exercise prompt. Let $S \sim \text{Bin}(4, \theta)$. Test $H_0 : \theta = 1/2$ against $H_1 : \theta = 3/4$ by rejecting when $S = 4$. Find the size and the Type II error probability. See Exercise 1 in Section 6 and its complete solution.

2 An example of normal distribution

High priority *An exact Normal mean test*

Following BD Example 4.1.4, pp. 218–219, let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ with *known* $\sigma > 0$. Fix $\mu_1 > \mu_0$ and consider the two simple hypotheses $H_0 : \mu = \mu_0$ versus $H_1 : \mu = \mu_1$. Write Φ for the standard Normal cdf and $z_q = \Phi^{-1}(q)$. Define

$$Z = \frac{\sqrt{n}(\overline{X} - \mu_0)}{\sigma}, \qquad d = \frac{\sqrt{n}(\mu - \mu_0)}{\sigma}.$$

Under H_0, $Z \sim N(0,1)$; under H_1, $Z \sim N(d_1, 1)$ with $d_1 = \sqrt{n}(\mu_1 - \mu_0)/\sigma > 0$. More generally, the rejection probability of $\varphi_c = \mathbb{1}\{Z \geq c\}$ at a given μ is

$$\pi_c(\mu) = 1 - \Phi(c - d). \tag{1}$$

In particular, the Type I error probability is $1 - \Phi(c)$ and power is $1 - \Phi(c - d_1)$. Thus $c = z_{1-\alpha}$ gives size exactly α, and we reject when

$$\overline{X} \geq \mu_0 + z_{1-\alpha}\frac{\sigma}{\sqrt{n}}. \tag{2}$$

Writing the fixed alternative as $\mu_1 = \mu_0 + \Delta$, $\Delta > 0$, power is $1 - \Phi(z_{1-\alpha} - \sqrt{n}\Delta/\sigma)$. More observations increase this power. For fixed n, however, power tends to α as $\Delta \downarrow 0$: arbitrarily small departures remain difficult to detect.

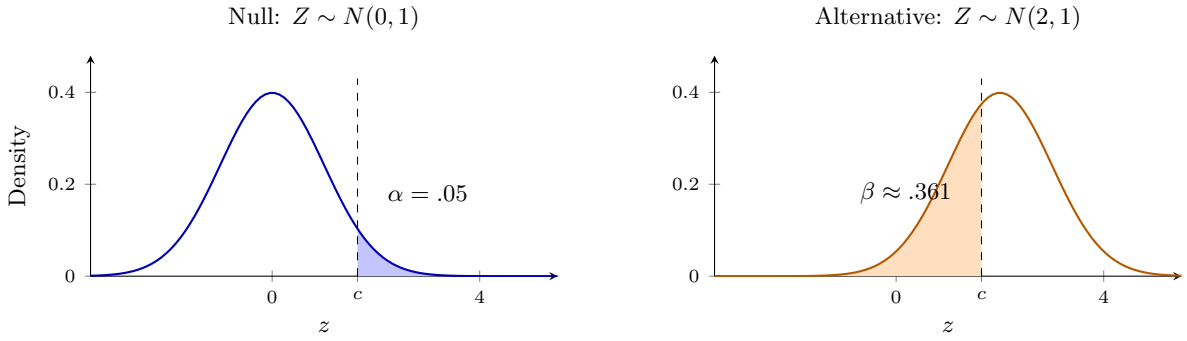


Figure 1: The same cutoff $c = z_{.95} \approx 1.645$ is used in both panels. The left shaded area is Type I error under the simple null; the right shaded area is Type II error at standardized effect $d = 2$. Power there is about 0.639. Original course drawing of the Normal calculation in BD Example 4.1.4.

Exercise prompt. In this model, take $\alpha = .05$, $\sigma = 2$, and target departure $\Delta = 1$. What is the smallest integer n giving power at least .80? Use $z_{.95} \approx 1.6449$ and $z_{.80} \approx .8416$. See Exercise 2 and its complete solution.

Mid priority *An exact binomial test can be conservative*

BD Example 4.1.3, pp. 215–218, uses $S \sim \text{Bin}(10, \theta)$ and reference probability $\theta_0 = .3$. Start with the simple hypotheses $H_0 : \theta = .3$ and $H_1 : \theta = .5$, rejecting for $S \geq k$. For any specified θ, the rejection probability is

$$\pi_k(\theta) = \sum_{j=k}^{10} \binom{10}{j} \theta^j (1-\theta)^{10-j}.$$

Direct summation gives

$$P_{.3}(S \geq 5) \approx .15027, \qquad P_{.3}(S \geq 6) \approx .04735.$$

The smallest integer cutoff for level .05 is $k = 6$. Its size is .04735, and its power at $\theta = .5$ is only $P_{.5}(S \geq 6) = 386/1024 \approx .37695$. The unused Type I error budget comes from the discrete jumps in the tail probabilities. "Exact" means calibrated using the actual sampling law; it does not mean that the size equals the nominal level.

High priority *Exact versus asymptotic calibration*

The Normal calculation above is exact for every n under its Normal model. For general i.i.d. observations with mean μ, finite known variance $\sigma^2 > 0$, and a fixed distribution, the CLT only gives $\sqrt{n}(\overline{X} - \mu)/\sigma \Rightarrow N(0, 1)$. Using Normal critical values then gives a large-sample approximation, not a finite-sample error guarantee. We return to estimated standard errors and approximate confidence intervals in Lecture 22.

3 p-values

High priority *A tail probability for the observed statistic*

Suppose the observed standardized mean is $z_{\text{obs}} = 2$ and the specified alternative is a positive shift. Under the simple null $\mu = \mu_0$, $Z \sim N(0, 1)$. The question is: *if this null model generated another sample, how often would its standardized mean be at least 2?* The answer is $P\{Z \geq 2\} \approx .02275$. This is the observed upper-sided p-value. The observed threshold 2 is fixed in this calculation; the hypothetical new sample is random. We count the whole tail, not the probability of obtaining exactly 2 (which is zero in this continuous model).

More generally, choose a statistic T for which larger values count against H_0, before looking at the data. For a simple null P_0, write

$$p(x) = P_0\{T(X') \geq T(x)\}. \tag{3}$$

Here x is the observed data and X' denotes a fresh sample from the null model. "At least as extreme" means $T(X') \geq T(x)$ according to this *chosen* statistic. For the fixed positive Normal shift in Section 2, use $T = Z$; for a fixed negative shift, use $T = -Z$. The statistic is selected to detect the specified alternative. There is no direction-free notion of extremeness. Two-sided alternatives are developed in Section 5.

Thus $p(x)$ is the size of the tail region starting at the observed statistic. It is the smallest size among the upper-tail tests that reject this observation (BD Proposition 4.1.2, p. 222). The corresponding level-α decision is to reject when $p(X) \leq \alpha$. In a discrete model, include the observed value in the tail: replacing $\geq$ by $>$ can invalidate the calibration.

The significance level and the p-value have different jobs. The level α is the error tolerance chosen for the testing procedure; the p-value is calculated from the data. With the same upper-sided test and $z_{\text{obs}} = 2$, different prespecified levels would give:

Level α	Reject when $Z \geq z_{1-\alpha}$	Decision at $Z = 2$
.10	$Z \geq 1.282$	Reject
.05	$Z \geq 1.645$	Reject
.025	$Z \geq 1.960$	Reject
.01	$Z \geq 2.326$	Do not reject

The p-value .02275 records where the decision switches as the level varies: this observation is rejected precisely when $\alpha \geq p(x)$ for this family. It is useful to report the p-value even when no single cutoff is specified. Choosing a favorable α after seeing the result does not give a test with a prespecified error guarantee.

Definition 3 (Validity). *A statistic $p(X) \in [0,1]$ is a **valid p-value** for H_0 if, for every $u \in [0,1]$,*

$$P_0\{p(X) \le u\} \le u. \tag{4}$$

This inequality immediately guarantees that rejecting at $p \le \alpha$ controls Type I error at level α. For example, if a null model is true, at most 5% of repeated samples produce $p(X) \le .05$. This statement concerns the random p-value across experiments; the observed value .02275 is one realization of that statistic. Write $F_0(t) = P_0\{T(X) \le t\}$ for the null CDF of the statistic. For a simple null and a continuously distributed statistic, $p(X) = 1 - F_0(T(X))$ is exactly uniform on $[0,1]$ by the probability integral transform (BD, p. 222 and Problem 4.1.5). For discrete statistics it is generally conservative. Exercise 5 gives a proof; Section 5 explains the extension to a composite null.

Normal examples. For the simple hypotheses in Section 2, $p_+(x) = 1 - \Phi(z_{\text{obs}})$. Taking $n = 25$, $\sigma = 2$, and $\overline{x} - \mu_0 = .8$ gives $z_{\text{obs}} = 2$ and $p_+(x) \approx .02275$. If the specified alternative has $\mu_1 < \mu_0$, use the lower tail $p_-(x) = \Phi(z_{\text{obs}})$ instead. The direction must be prespecified.

For the binomial example, observing $s = 6$ gives $p(s) = \sum_{j=6}^{10} \binom{10}{j} (.3)^j (.7)^{10-j} \approx .04735$. No Normal approximation is needed.

High priority *What a p-value does and does not say*

A small p-value says that the observed statistic lies in a tail that has small probability under P_0, using the chosen ordering of outcomes. It is not the probability that H_0 is true, and it is not the probability that this particular rejection is mistaken. A large p-value does not establish H_0: the test may have little power against a relevant alternative. In the example $p = .02275$ means that, under the simple null model, about 2.275% of repeated samples have $Z \ge 2$. It does *not* mean that the null has a 2.275% chance of being true, or that the alternative has a 97.725% chance of being true. Such probabilities would require an additional Bayesian model, including a prior distribution.

Statistical significance also does not measure effect size or scientific importance. Report the estimate, its uncertainty, the model, and how the analysis was chosen. Searching among many analyses and reporting only the smallest unadjusted p-value changes the error calculation. These interpretation points follow the ASA's 2016 principles [1]; they complement the testing framework rather than replace its assumptions.

Instructor's Note *One observation, many possible repeats*

Imagine replaying the experiment while the null model stays fixed. The p-value asks how often that replay would produce a test statistic at least as far in the chosen direction as the one we saw. The replay changes the data; it does not randomly switch the null hypothesis between true and false.

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4 Why likelihood ratios give powerful tests

Mid priority *Most powerful tests and randomization*

For simple hypotheses $H_0 : P = P_0$ and $H_1 : P = P_1$, let f_0, f_1 be densities with respect to the same measure ν (counting measure for a discrete model, Lebesgue measure for a continuous one). The simple likelihood ratio is $L(x) = f_1(x)/f_0(x)$, with $L = \infty$ when $f_0 = 0 < f_1$. Large values favor the specified alternative over the specified null.

A **randomized test** allows $\varphi(x) \in [0,1]$: given x, reject with probability $\varphi(x)$ using an independent randomizer. Then its rejection probability is $E_\theta \varphi(X)$. Ordinary tests take only the values zero and one. Randomization lets us fill the gaps between attainable sizes in discrete models.

A test of level α is **most powerful** (MP) against P_1 if its power is at least that of every other level-α test, including randomized tests.

Mid priority *The Neyman–Pearson argument*

Theorem 1 (BD Theorem 4.2.1(a), pp. 224–225, restated)**.** *Suppose* $0 < \alpha < 1$ *and there are* $k \geq 0$ *finite and a test* φ^* *with* $E_0\varphi^* = \alpha$ *such that*

$$\varphi^*(x) = \begin{cases} 1, & f_1(x) > kf_0(x), \\ 0, & f_1(x) < kf_0(x), \\ \gamma(x) \in [0,1], & f_1(x) = kf_0(x). \end{cases}$$

Then φ^* *is most powerful among tests with* $E_0\varphi \leq \alpha$.

Low priority *Proof (optional)*

Proof (the sign argument of BD, written with densities). For every competitor $0 \leq \varphi \leq 1$,

$$(\varphi^*(x) - \varphi(x))(f_1(x) - kf_0(x)) \geq 0.$$

Indeed, where the second factor is positive, $\varphi^* = 1$; where it is negative, $\varphi^* = 0$; and on equality the product vanishes. Integrating with respect to ν gives

$$E_1\varphi^* - E_1\varphi \geq k(E_0\varphi^* - E_0\varphi) = k(\alpha - E_0\varphi) \geq 0.$$

Thus the proposed rule has at least as much power as every competitor. The density form also handles points where $f_0 = 0$ without dividing by zero. □

Mid priority *Calibrating and interpreting the likelihood-ratio rule*

The equality rule must be calibrated so that the stated size condition holds. When the likelihood ratio has no null atom at the cutoff, no boundary randomization is needed. BD Theorem 4.2.1(b) also proves existence at every size when randomization is permitted; Exercise 6 illustrates the construction in our binomial example.

Instructor's Note *Where to spend the error budget*

Each region where we reject spends some probability under the null and gains some probability under the alternative. The likelihood ratio measures that gain relative to its cost. Spend the budget first on the observations with the largest ratios. The proof formalizes why swapping such observations for lower-ratio ones cannot improve power.

A four-outcome illustration. Suppose the entire observation is one of A, B, C, D, with null and alternative probabilities $(.4, .3, .2, .1)$ and $(.1, .2, .3, .4)$, respectively. The likelihood ratios are $1/4, 2/3, 3/2, 4$. At level $\alpha = .30$, rejecting on $\{C, D\}$ spends $.20 + .10 = .30$ under the null and gains $.30 + .40 = .70$ under the alternative. Rejecting only on B has the same size but power .20. The first rule includes exactly the outcomes with $f_1/f_0 > 1$; Theorem 1 says it is MP even among randomized competitors. The choice is governed by the ratio, not by f_1 alone. This original numerical example illustrates BD Theorem 4.2.1; the error-budget explanation is also given in the Berkeley Stat 210A reader, Section 3.1 [4].

Mid priority *The Normal test is most powerful for a fixed alternative*

For i.i.d. $N(\mu, \sigma^2)$ observations with known σ^2, compare $\mu = \mu_0$ with a fixed $\mu_1 > \mu_0$. Expanding the squares in the densities,

$$\log \frac{f_{\mu_1}(x)}{f_{\mu_0}(x)} = \frac{\mu_1 - \mu_0}{\sigma^2} \sum_{i=1}^{n} x_i - \frac{n(\mu_1^2 - \mu_0^2)}{2\sigma^2}. \tag{5}$$

It is strictly increasing in $\bar{x}$. The rule in (2) is therefore MP at level α for this alternative (BD Example 4.2.1, pp. 225–226).

Exercise prompt. Let X_i be i.i.d. Bernoulli(θ), $S = \sum_i X_i$, and $0 < \theta_0 < \theta_1 < 1$. Show that $f_{\theta_1}(x)/f_{\theta_0}(x)$ is increasing in S. What rejection direction does the Neyman–Pearson lemma suggest? See Exercise 3 and its complete solution.

5 Generalized likelihood ratios

So far we have compared two specified distributions. We now allow a hypothesis to contain several distributions. We first extend error control and p-values, then ask when one test can be optimal for many alternatives, and finally generalize the likelihood-ratio construction.

5.1 Composite hypotheses and error control

High priority *From two distributions to two sets of distributions*

Let $X \sim P_\theta$ in a model $\{P_\theta : \theta \in \Theta\}$, and partition Θ into disjoint, nonempty sets Θ_0, Θ_1. We test

$$H_0 : \theta \in \Theta_0 \qquad \text{against} \qquad H_1 : \theta \in \Theta_1.$$

Each hypothesis is simple or composite according to the number of data distributions it permits. For Normal data with known variance, $H_0 : \mu = \mu_0$ is simple even when $H_1 : \mu \neq \mu_0$ is composite. The hypotheses $\mu \leq \mu_0$ and $\mu > \mu_0$ are both composite. If the variance is unknown, even $\mu = \mu_0$ is composite: different variances give different distributions. The variance is then a **nuisance parameter**, needed to specify the model but not targeted by the question.

Definition 4 (General power function and size; BD Definition 4.1.1, p. 217)**.** *For an ordinary test, define*

$$\pi_\varphi(\theta) = P_\theta\{\varphi(X) = 1\}, \qquad \text{size}(\varphi) = \sup_{\theta \in \Theta_0} \pi_\varphi(\theta).$$

A level-α test has size at most α. At an alternative parameter, power is $\pi_\varphi(\theta)$ and Type II error probability is $1 - \pi_\varphi(\theta)$. For a randomized test, replace the rejection probability by $E_\theta\varphi(X)$.

For a composite null, controlling error at one convenient parameter is insufficient: the bound must hold at *every* null parameter. Power can likewise differ across alternative parameters; there need not be one single Type II error probability. For approximate tests, pointwise convergence for each fixed distribution also does not by itself give uniform control over a class of null distributions.

Extend the Normal test. The rejection probability in (1) is increasing in μ. Consequently, for the enlarged null $\mu \leq \mu_0$,

$$\sup_{\mu \leq \mu_0} \pi_c(\mu) = \pi_c(\mu_0) = 1 - \Phi(c).$$

Thus the same cutoff $c = z_{1-\alpha}$ controls the entire composite null. This monotonicity argument is the extra step that the simple-null calculation did not require.

Mid priority *Extend the binomial test*

Enlarge the binomial hypotheses from Section 2 to $\theta \le .3$ versus $\theta > .3$. We must now check $P_\theta(S \ge k) \le \alpha$ for every $\theta \le .3$. To see that $P_\theta(S \ge k)$ increases with θ, use the same independent $U_i \sim \text{Unif}(0,1)$ for all parameter values and set $S_\theta = \sum_{i=1}^{10} \mathbb{1}\{U_i \le \theta\}$. Then S_θ is nondecreasing for every fixed set of uniforms, so its upper-tail probabilities are nondecreasing. Size is therefore attained at .3.

The rule $S \ge 6$ therefore has size .04735 under the enlarged null as well. Its power varies with the alternative θ.

5.2 p-values for a composite null

High priority *Protect against every null distribution*

If the null is composite, it permits several sampling distributions. To protect against false rejection under all of them, use the largest tail probability:

$$p(x) = \sup_{\theta \in \Theta_0} P_\theta\{T(X') \ge T(x)\}, \tag{6}$$

where the law of X' in each probability is P_θ. Assume the displayed supremum is a measurable function of x. This is a worst-case probability over null models, not an average over possible parameter values.

Validity now requires

$$P_\theta\{p(X) \le u\} \le u \quad \text{for every } \theta \in \Theta_0, \quad 0 \le u \le 1. \tag{7}$$

For any fixed null parameter, the supremum p-value is at least its simple-null tail p-value. Its event $\{p(X) \le u\}$ is therefore contained in the corresponding simple-null event, which has probability at most u. This proves validity, with details in Exercise 5. The size-of-the-tail interpretation in Section 3 extends by replacing one null probability with its supremum (BD Proposition 4.1.2).

For the Normal upper-tail statistic and $\mu \le \mu_0$, the largest tail probability occurs at μ_0, so $p_+(x) = 1 - \Phi(z_{\text{obs}})$ remains valid. For the enlarged binomial null, the same argument gives $p(s) = P_{.3}(S \ge s)$. These formulas stay the same because we checked where their worst-case null probabilities occur.

5.3 From most powerful to uniformly most powerful

Mid priority *One rule for a whole family of alternatives*

Section 4 proved that the Normal upper-tail rule is MP against each specified positive shift. We can now ask whether the same rule solves the composite problem.

Definition 5 (BD Definition 4.3.1, p. 227)**.** *A level-α test is **uniformly most powerful** (UMP) if it is most powerful at every parameter value in the alternative.*

Our upper-tail rule is the same for every $\mu_1 > \mu_0$. It has level α over the composite null $\mu \le \mu_0$, as already checked. Every competitor valid over that composite null is also valid at μ_0, so the Neyman–Pearson comparison applies to it at each $\mu_1 > \mu_0$. This proves that the Normal upper-tail test is UMP for $\mu \le \mu_0$ versus $\mu > \mu_0$.

This conclusion does not automatically extend to a two-sided alternative. For a positive Normal shift the MP rule rejects in the upper tail; for a negative shift it rejects in the lower tail. Because the simple likelihood ratio is strictly monotone and continuously distributed, the uniqueness part of BD Theorem 4.2.1(c) forces these rules up to null sets. No single rule can equal both when $0 < \alpha < 1$. Thus there is no UMP test against all $\mu \ne \mu_0$ among unrestricted level-α tests.

5.4 Compare the best null and full-model fits

Mid priority *Definition and calibration*

With a composite alternative there is no single alternative density to put in the numerator of a simple likelihood ratio. A general construction is

$$\Lambda(x) = \frac{\sup_{\theta\in\Theta_0} f_\theta(x)}{\sup_{\theta\in\Theta} f_\theta(x)}, \qquad W(x) = -2\log\Lambda(x), \tag{8}$$

for data where the denominator is positive and finite. Since $\Theta_0 \subset \Theta$, $0 \le \Lambda \le 1$. Reject for small Λ, equivalently large W; set $W = \infty$ if $\Lambda = 0$. When the suprema are attained, W is twice the difference between the unrestricted and constrained maximized log likelihoods (Rigollet, slides 23–24 [3]). This ratio uses *null over full-model fit*, the opposite direction from f_1/f_0 above. A null distribution or justified approximation is still needed to choose a critical value. The construction alone does not prove level control or UMP optimality.

Two-sided Normal test, derived directly. For known σ^2 and $H_0 : \mu = \mu_0$ versus $H_1 : \mu \ne \mu_0$, the unrestricted MLE is $\widehat{\mu} = \overline{x}$. The identity

$$\sum_{i=1}^n (x_i - \mu_0)^2 = \sum_{i=1}^n (x_i - \overline{x})^2 + n(\overline{x} - \mu_0)^2$$

gives

$$\Lambda(x) = \exp\left\{-\frac{n(\overline{x}-\mu_0)^2}{2\sigma^2}\right\}, \qquad W = \frac{n(\overline{X}-\mu_0)^2}{\sigma^2} = Z^2.$$

Under H_0, $W \sim \chi^2_1$ *exactly*, since it is the square of one standard Normal variable. Rejecting for $W \ge z^2_{1-\alpha/2}$ is precisely the two-sided Z test. This example does not need a large-sample likelihood theorem.

One rejection rule, a family of alternatives. Figure 2 extends Figure 1 to this two-sided test. Write $d = \sqrt{n}(\mu - \mu_0)/\sigma$ and $c = z_{1-\alpha/2}$. The statistic has distribution $Z \sim N(d, 1)$; the null fixes $d = 0$, while the composite alternative allows every $d \ne 0$. The rejection region $\{|Z| \ge c\}$ stays the same as d changes. Under the null its two tails each have probability $\alpha/2$. At any alternative $d \ne 0$, the unshaded tails give power and the shaded central region gives Type II error:

$$\beta(d) = P_d(-c < Z < c) = \Phi(c - d) - \Phi(-c - d), \qquad \pi(d) = 1 - \beta(d).$$

For $\alpha = .05$ and $d = \pm 2$, $\beta(d) \approx .484$ and power is about .516. The two alternative panels are representatives of the family; there is no single sampling distribution under the composite H_1.

High priority *Two-sided p-values*

The null here is still simple; the alternative includes departures in both directions. The GLR calculation selects $|Z|$ (equivalently Z^2) as the statistic measuring disagreement with the null. For $H_0 : \mu = \mu_0$ versus $H_1 : \mu \ne \mu_0$, use $T = |Z|$ instead:

$$p_{\text{two}}(x) = P\{|N(0,1)| \ge |z_{\text{obs}}|\} = 2\{1 - \Phi(|z_{\text{obs}}|)\}.$$

The two-sided test rejects for $|Z| \ge z_{1-\alpha/2}$. For example, $n = 25$, $\sigma = 2$, and $\overline{x} - \mu_0 = .8$ give $z_{\text{obs}} = 2$. The one-sided p-value is about .02275 and the two-sided p-value about .04550. A p-value depends on which departures the test was designed to detect. The two-sided calculation counts both $Z \ge 2$ and $Z \le -2$, because equally large departures in either direction matter.

Exercise prompt. Under the known-variance Normal model, an observation gives $z_{\text{obs}} = -2$. Find p-values for a prespecified upper-sided, lower-sided, and two-sided test. What is the size of the rule that chooses the favorable one-sided direction after seeing the sign of Z, and rejects if that one-sided p-value is at most .05? See Exercise 4 and its complete solution.

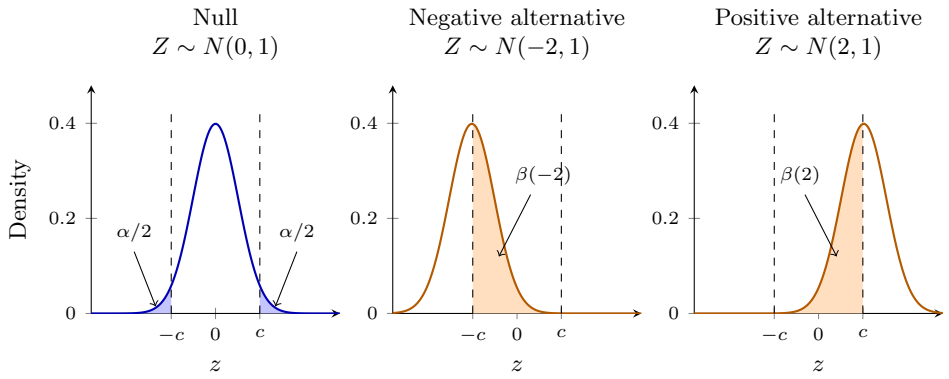


Figure 2: The two-sided Normal GLR test uses the same cutoffs $\pm c$, where $c = z_{.975} \approx 1.960$, for every distribution in the model. Blue shading shows the null's two error tails (total .05); orange shading shows Type II error for two representative alternatives. Course adaptation of the rejection-region diagram in UC Berkeley's Stat 210A reader, Section 2.2 [4], licensed under CC BY 4.0. Redrawn as three panels at $\alpha = .05$, with $d = \pm 2$ and the lecture's β-for-Type-II-error convention.

Mid priority *A numerical generalized likelihood ratio*

Compute the two best fits. Return to $n = 25$, $\sigma = 2$, and now set $\mu_0 = 0$ and $\overline{x} = .8$. For these fixed data, write $L_n(\mu) = f_\mu(x)$ for the sample likelihood, as in Lecture 22. This is a function of the parameter; the earlier $L(x) = f_1(x)/f_0(x)$ is a simple likelihood ratio, a function of the data. Divide the sample likelihood by its maximum:

$$\frac{L_n(\mu)}{L_n(\widehat{\mu})} = \exp\left\{-\frac{25}{8}(\mu - .8)^2\right\}, \qquad \widehat{\mu} = \overline{x} = .8.$$

The full model is maximized at .8; the null restricts μ to 0. The ratio of these two fits is $\Lambda = e^{-2} \approx .1353$ and twice the log-likelihood gap is $W = 4$. The data stay fixed while the likelihood compares parameter values; it is not a probability distribution for μ.

To decide whether this gap is unusually large, return to the sampling distribution: $P_{\mu_0}\{\chi^2_1 \geq 4\} = 2\{1 - \Phi(2)\} \approx .04550$. The likelihood comparison defines the statistic; its null distribution supplies the p-value.

High priority *From testing a value to reporting a set of values*

Instead of testing only one μ_0, imagine repeating the same two-sided test for each candidate value m. The values that are not rejected form

$$\left\{m : |\overline{X} - m| < z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right\}.$$

This is the familiar Normal interval, here with open endpoints because the test rejects at equality. Closing the endpoints does not change its coverage in this continuous model. Lecture 22 will define coverage formally, prove the general test–confidence-set duality, and develop pivotal and approximate intervals (BD Sections 4.4–4.5). Testing and interval estimation can thus be two views of the same sampling calculation.

Key ideas

- A level-α rule controls false rejection for every null parameter; its attained size can be smaller than α.
- Power depends on the alternative, effect size, and sample size. Nonrejection need not mean high support for the null.
- A valid p-value is calibrated by $P_\theta(p \le u) \le u$ under the null; it is not a posterior probability.
- Neyman–Pearson gives optimality for simple hypotheses; UMP requires the same rule to work across alternatives.
- A generalized likelihood ratio compares constrained and unrestricted fits and still needs calibration.

Looking ahead

The next lecture will make the interval interpretation precise: what is random, what coverage guarantees, and how pivots and tests produce confidence sets. We will then use the earlier CLT and Slutsky tools when exact sampling distributions are unavailable.

Supplemental reference: key notions and definitions

This table links back to the definitions and results in the lecture. Each row keeps the priority of the corresponding material. The Neyman–Pearson proof is low priority by the instructor's direction; all other rows retain the priority of their corresponding lecture blocks.

Notion	Definition or key fact	Tier
Null and alternative	Start with P_0 versus P_1; more generally test $\theta \in \Theta_0$ versus $\theta \in \Theta_1$. Control false rejection over the null.	High
Simple and composite hypotheses	A simple hypothesis specifies one data distribution; a composite hypothesis allows several. A nuisance parameter can make a point restriction composite.	High
Test and rejection region	A statistic summarizes the data; a rejection region R specifies when to reject. For an ordinary test, the decision is $\mathbf{1}\{X \in R\}$.	High
Type I and Type II errors	Type I: reject at a null parameter. Type II: do not reject at an alternative; its probability is $1 - \pi(\theta)$ for $\theta \in \Theta_1$.	High
Power function	$\pi(\theta) = P_\theta(X \in R)$ on the whole parameter space; under an alternative, it is the probability of detecting the departure.	High
Size and level	Simple null: size is $P_0(R)$. Composite null: it is $\sup_{\theta \in \Theta_0} \pi(\theta)$. A level-$\alpha$ test has size at most α, possibly strictly smaller.	High
Level versus p-value	α is the chosen error tolerance; $p(x)$ is computed from the observed data. The calibrated decision rule rejects when $p(X) \le \alpha$.	High
Tail p-value	For a simple null, $p(x) = P_0\{T(X') \ge T(x)\}$. For a composite null, take the supremum of these probabilities over Θ_0. Fix the observed x; the repeat sample X' is random.	High
Valid p-value	$P_\theta\{p(X) \le u\} \le u$ for every null parameter and every $u \in [0,1]$. This is a repeated-sample error guarantee, not a probability that H_0 is true.	High
Known-variance Normal test	$Z = \sqrt{n}(\overline{X} - \mu_0)/\sigma$ is $N(0,1)$ at $\mu = \mu_0$. The upper-sided level-α rule rejects for $Z \ge z_{1-\alpha}$.	High
One-sided and two-sided p-values	For a Normal observation z, the upper-sided p-value is $1 - \Phi(z)$; the two-sided value is $2\{1 - \Phi(\lvert z \rvert)\}$. Choose the direction before the data.	High

Notion	Definition or key fact	Tier
Exact and asymptotic calibration	Exact calibration uses a finite-sample null law. Asymptotic calibration uses a limiting law and need not control error exactly at a given n.	High
Randomized test	$\varphi(x) \in [0,1]$ is the conditional probability of rejection given x; power is $E_\theta \varphi(X)$. It can fill gaps between discrete attainable sizes.	Mid
Most powerful (MP)	An MP level-α test maximizes power against a specified alternative among all level-α tests, including randomized ones.	Mid
Neyman–Pearson lemma	For simple hypotheses, a rule rejecting when $f_1 > kf_0$, calibrated to size α with possible randomization at equality, is MP.	Mid
Neyman–Pearson proof	Integrate $(\varphi^* - \varphi)(f_1 - kf_0) \geq 0$ and use the size constraint. The full proof is optional reading.	Low
Uniformly most powerful (UMP)	One level-α test is MP at every alternative parameter. The known-variance Normal upper-tail rule is UMP for $\mu \leq \mu_0$ versus $\mu > \mu_0$.	Mid
Generalized likelihood ratio	$\Lambda = \sup_{\Theta_0} f_\theta / \sup_\Theta f_\theta$ compares null and full-model fits. Reject for small Λ, or large $W = -2\log\Lambda$. A null calibration is still required.	Mid
Test inversion (preview)	Collect the parameter values not rejected by the corresponding tests. For the two-sided Normal test this gives the familiar interval around $\overline{X}$; coverage is developed in Lecture 22.	High

6 Exercises and complete solutions

High priority *Errors for a small binomial experiment*

Exercise 1 (Original practice using BD Section 4.1). Let $S \sim \text{Bin}(4, \theta)$. First test $H_0 : \theta = 1/2$ against $H_1 : \theta = 3/4$ by rejecting when $S = 4$. Find its size, determine whether it has level .05, and find its Type II error probability. After Section 5, extend the null to $\theta \leq 1/2$ and the alternative to $\theta > 1/2$. Show that the size is unchanged.

Solution. For the simple null, size is $P_{1/2}(S = 4) = (1/2)^4 = 1/16 = .0625$, so the rule is not level .05. Under the simple alternative, power is $(3/4)^4 = 81/256$ and Type II error is $175/256 \approx .6836$. For the composite extension, the power function is $\pi(\theta) = P_\theta(S = 4) = \theta^4$. It is increasing on $[0,1]$, so $\sup_{\theta \leq 1/2} \theta^4 = 1/16$ and the size is unchanged. The two error probabilities are evaluated at different parameter values; they need not sum to one. □

High priority *Choose a sample size for a specified effect*

Exercise 2 (Adapted from BD Example 4.2.1, p. 226; new numerical values). For i.i.d. $N(\mu, \sigma^2)$ data with known σ, test $\mu = \mu_0$ against the specified $\mu_1 = \mu_0 + \Delta$ at level α, where $\Delta > 0$ and $0 < \beta_0 < 1 - \alpha$. Derive the sample size needed for power at least $1 - \beta_0$ at μ_1. The same calculation applies to the composite extension in Section 5. Evaluate it for $\alpha = .05$, $\beta_0 = .20$, $\sigma = 2$, and $\Delta = 1$.

Solution. Using (1), the desired inequality is

$$1 - \Phi\left(z_{1-\alpha} - \frac{\sqrt{n}\Delta}{\sigma}\right) \geq 1 - \beta_0.$$

As Φ is increasing, this is equivalent to

$$\frac{\sqrt{n}\Delta}{\sigma} \geq z_{1-\alpha} - z_{\beta_0} = z_{1-\alpha} + z_{1-\beta_0}.$$

The right side is positive under $\beta_0 < 1 - \alpha$, so squaring is valid:

$$n \geq \frac{\sigma^2}{\Delta^2}(z_{1-\alpha} + z_{1-\beta_0})^2.$$

Here the bound is $4(1.6449 + .8416)^2 \approx 24.73$, hence $n = 25$. The exact powers are about .789 at $n = 24$ and .804 at $n = 25$. This design targets the specified departure $\Delta = 1$, not every positive departure. □

Mid priority *Why large binomial counts favor the alternative*

Exercise 3 (Adapted from BD equation (4.2.1), p. 223). Let X_i be i.i.d. Bernoulli(θ), $S = \sum_{i=1}^n X_i$, and $0 < \theta_0 < \theta_1 < 1$. Show that the simple likelihood ratio increases with S and describe the Neyman–Pearson rejection rule. Explain the role of boundary randomization.

Solution. For an observed binary vector x with sum s,

$$\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} = \left(\frac{\theta_1}{\theta_0}\right)^s \left(\frac{1-\theta_1}{1-\theta_0}\right)^{n-s} = \left(\frac{1-\theta_1}{1-\theta_0}\right)^n \left(\frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)}\right)^s.$$

The last base exceeds one because $\theta_1 > \theta_0$. The same ratio results from the binomial pmf since $\binom{n}{s}$ cancels. Thus reject for large counts. To attain an arbitrary prescribed size, choose an integer b with $P_{\theta_0}(S > b) \leq \alpha \leq P_{\theta_0}(S \geq b)$, reject for $S > b$, and randomize at $S = b$ with probability

$$\gamma = \frac{\alpha - P_{\theta_0}(S > b)}{P_{\theta_0}(S = b)} \in [0, 1].$$

This rule has size α at θ_0 and is MP by Theorem 1. For a nonrandomized rule, choose an attainable upper-tail size not exceeding α; it may be strictly smaller than α. □

High priority *Choose the direction before seeing the data*

Exercise 4 (Original practice using BD's Normal p-value example). For the known-variance Normal model, an observation gives $z_{\text{obs}} = -2$. Find the p-values for the prespecified alternatives $\mu > \mu_0$, $\mu < \mu_0$, and $\mu \neq \mu_0$. Compute the Type I error under $\mu = \mu_0$ if we instead choose the favorable one-sided direction after observing Z and reject when that one-sided p-value is at most .05.

Solution. For the upper alternative, $p_+ = 1 - \Phi(-2) = \Phi(2) \approx .97725$. For the lower alternative, $p_- = \Phi(-2) \approx .02275$. For the two-sided alternative, $p_{\text{two}} = 2\Phi(-2) \approx .04550$. The data point in the opposite direction from the upper alternative.

After choosing the favorable sign, the reported value is $\min(p_+, p_-) = 1 - \Phi(|Z|)$. The rule rejects when $|Z| \geq z_{.95}$. Under the null,

$$P_{\mu_0}(|Z| \geq z_{.95}) = 2(1 - \Phi(z_{.95})) = .10.$$

It therefore has size .10, not .05. For a level-.05 two-sided test, use the two-sided p-value, or equivalently $|Z| \geq z_{.975}$. □

Low priority *Why the tail p-value is valid*

Exercise 5 (BD Problem 4.1.5, p. 270, with discrete and composite extensions). For a real-valued statistic T under a simple null P_0, put $G(t) = P_0(T \geq t)$ and $p = G(T)$. Prove $P_0(p \leq u) \leq u$. Show equality when T has a continuous distribution. Then prove validity of the supremum p-value in (6). The first inequality and composite extension are added course exercises.

Solution. Fix $0 < u < 1$. Because G is nonincreasing, $A_u = \{t : G(t) \le u\}$ is an upper set. If empty its probability is zero. Otherwise it is an upper ray (and it cannot be all of $\mathbb{R}$, since $G(t) \to 1$ as $t \to -\infty$). If $A_u = [a, \infty)$, then $P_0(T \in A_u) = G(a) \le u$. If $A_u = (a, \infty)$, take $t_j \downarrow a$ with $t_j \in A_u$. The events $\{T \ge t_j\}$ increase to $\{T > a\}$, and each has probability at most u. Continuity of probability gives the same bound. Thus $P_0\{G(T) \le u\} \le u$. The case $u = 1$ is immediate; the case $u = 0$ follows by taking $u \downarrow 0$.

If F is the continuous cdf of T, then $G(T) = 1 - F(T)$ and $F(T)$ is uniform. Here is a justification even if F has flat portions. For $0 < v < 1$, let $b = \sup\{t : F(t) \le v\}$. Continuity and the cdf limits imply $F(b) = v$ and $\{t : F(t) \le v\} = (-\infty, b]$. Therefore $P_0(F(T) \le v) = F(b) = v$. Consequently $P_0(1 - F(T) \le u) = u$. Atoms prevent this equality in general, but not the bound.

For a composite null, fix any true $\theta \in \Theta_0$ and set $G_\theta(t) = P_\theta\{T(X') \ge t\}$. Pointwise $p(X) \ge G_\theta(T(X))$, so

$$P_\theta(p(X) \le u) \le P_\theta(G_\theta(T(X)) \le u) \le u.$$

Since θ was arbitrary, this proves validity over the whole null. It does not require a single parameter to attain the supremum for every observation. □

Low priority *Fill the gap between two binomial cutoffs*

Exercise 6 (Adapted from BD Section 4.2, p. 224). Let $S \sim \mathrm{Bin}(10, \theta)$ and test $\theta \le .3$ against $\theta > .3$. Reject for $S \ge 6$, reject with probability γ when $S = 5$, and otherwise do not reject. Choose γ to give size exactly .05 and explain why control at .3 controls the whole composite null.

Solution. At $\theta = .3$, rejection probability is

$$P_{.3}(S \ge 6) + \gamma P_{.3}(S = 5) = .0473489874 + \gamma(.1029193452).$$

Thus

$$\gamma = \frac{.05 - .0473489874}{.1029193452} \approx .0257582.$$

The values are obtained directly from the binomial pmf; using rounded intermediate tails changes the last digits. The rejection function is nondecreasing in s because $0 \le \gamma \le 1$. Coupling the counts using the same uniforms as in Section 5 shows that its expectation is nondecreasing in θ. Its maximum on $\theta \le .3$ therefore occurs at .3 and equals .05. This is a theoretical way to attain the exact budget. The simpler nonrandomized rule $S \ge 6$ remains a valid level-.05 test. □

Reference guide. BD Section 4.1 supplies the testing vocabulary, binomial and Normal examples, and Proposition 4.1.2 on p-values; Theorem 4.2.1 supplies the Neyman–Pearson lemma and its proof; Definition 4.3.1 and Example 4.2.1 explain UMP testing. The generalized likelihood-ratio formulation is supplemented by MIT's notes; the Normal derivation here is explicit. The four-outcome numerical example also credits Berkeley's error-budget explanation. Figure 1 is the original course Normal error diagram; the three-panel Figure 2 adapts Berkeley's Section 2.2 diagram to explain the composite alternative. The final two exercises are optional technical complements. All lecture exercise solutions are supplied here.

References

[1] American Statistical Association. American statistical association releases statement on statistical significance and p-values. `https://www.amstat.org/asa/files/pdfs/p-valuestatement.pdf`, 2016. March 7. Official summary of the six principles; accessed October 27, 2026.

[2] Peter J. Bickel and Kjell A. Doksum. *Mathematical Statistics: Basic Ideas and Selected Topics, Volume I.* Prentice Hall, Upper Saddle River, NJ, 2 edition, 2001.

[3] Philippe Rigollet. Parametric hypothesis testing. MIT OpenCourseWare, 18.650 Statistics for Applications, Fall 2016, 2016. Slides 23–24, "Likelihood ratio test." https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/pages/lecture-slides/. Accessed October 27, 2026.

[4] UC Berkeley Stat 210A. Hypothesis testing and the Neyman–Pearson lemma. Fall 2024 course reader, Sections 2.2 and 3.1, 2024. Normal rejection-region diagram and error-budget interpretation of likelihood ratios. Diagram adapted under CC BY 4.0. https://stat210a.berkeley.edu/fall-2024/reader/hypothesis-testing.html. Accessed October 27, 2026.

What changed

October 27, 2026: lecture revised and approved for publication. Added the Part VII provenance statement, clearer p-value explanations and the supplemental reference table; renamed Sections 2, 3 and 5. Subsequently removed Figures 2 and 3 and reorganized the lecture to develop simple hypotheses first, then composite hypotheses in Section 5. Added a new three-panel sampling-density figure in Section 5.4, with Berkeley attribution, at the instructor's subsequent request. Typeset the p in p-value throughout the lecture. Theorem 1's complete proof remains low priority. The HTML edition uses shorter paragraphs and itemized explanations while retaining the full TeX content. Both HTML and PDF editions were approved for publication at that stage. October 28, 2026, approved joint-review corrections: explicitly define the null CDF and distinguish the sample likelihood from the simple likelihood ratio. Restore the MIT and Berkeley reading links in HTML. The instructor approved these review changes for publication in the consistent TeX, PDF and HTML editions.